

Number theory – 1

Congruences: Fermat, Euler and the Chinese remainder theorem

Homework

14 сентября 2026 года

Theoretical minimum. The Euclidean algorithm gives Bézout's identity $ax + by = \gcd(a, b)$, hence Euclid's lemma and unique factorisation. The Chinese remainder theorem says that for pairwise coprime n_i the map $\mathbb{Z}/n\mathbb{Z} \rightarrow \prod \mathbb{Z}/n_i\mathbb{Z}$ is a ring isomorphism, so every question about $\mathbb{Z}/n\mathbb{Z}$ splits into prime powers and φ is multiplicative. Euler's theorem $a^{\varphi(n)} \equiv 1 \pmod{n}$ for $\gcd(a, n) = 1$ follows because multiplication by a permutes the invertible residues; the same argument gives Wilson's theorem $(p-1)! \equiv -1 \pmod{p}$, which also characterises primes. Hensel's lemma lifts a root of f modulo p with $f'(a) \not\equiv 0$ to a root modulo every power of p , so obstructions live modulo p and at multiple roots. To disprove solvability in integers, find a modulus: squares are $0, 1 \pmod{4}$ and $0, 1, 4 \pmod{8}$, cubes are $0, \pm 1 \pmod{9}$.

1. Determine all positive integers n with the property that for every integer m coprime to n there is a permutation π of $\{1, \dots, n\}$ with $\pi(\pi(k)) \equiv mk \pmod{n}$ for all k . (Putnam 2023)
2. Call a square-free positive integer n *almost prime* if $n \mid x^{d_1} + x^{d_2} + \dots + x^{d_k} - kx$ for all integers x , where $1 = d_1 < d_2 < \dots < d_k = n$ are all the divisors of n . Find all almost primes. (IMC)
3. Does there exist an infinite set M of positive integers such that for any $a, b \in M$ with $a < b$ the sum $a + b$ is square-free? (IMC)
4. Let $p > 5$ be a prime and let $f(p)$ be the number of infinite sequences $a_1, a_2, \dots$ with $a_n \in \{1, 2, \dots, p-1\}$ and $a_n a_{n+2} \equiv 1 + a_{n+1} \pmod{p}$ for all $n \geq 1$. Prove that $f(p)$ is congruent to 0 or 2 modulo 5. (Putnam 2022)

5. Prove Hensel's lemma as stated, and use it to solve $x^2 \equiv 2 \pmod{7^3}$ and to show that $x^2 \equiv -1 \pmod{p^k}$ is solvable for every k whenever $p \equiv 1 \pmod{4}$. Then give an example where the hypothesis $f'(a) \not\equiv 0$ fails and the lifting breaks down.
6. Using the Chinese remainder theorem prove that for every n there exist n consecutive integers each of which is divisible by a perfect square greater than 1, and n consecutive integers none of which is a prime power.
7. Let p be an odd prime. Prove Wolstenholme's theorem in the weak form: the numerator of $1 + \frac{1}{2} + \cdots + \frac{1}{p-1}$ is divisible by p for $p \geq 3$, and by p^2 for $p \geq 5$.
8. Let $n \geq 1$ be odd. Determine all functions f from the integers to the integers such that $f(x) - f(y)$ divides $x^n - y^n$ for all integers x, y . (IMO shortlist 2011, N3)
9. Let f be a function from the integers to the positive integers such that $f(m) - f(n)$ is divisible by $f(m - n)$ for all integers m, n . Prove that for all m, n with $f(m) \leq f(n)$ the number $f(m)$ divides $f(n)$. (IMO shortlist 2011, N5)
10. Let x and y be positive integers. Prove that if $x^{2^n} - 1$ is divisible by $2^n y + 1$ for every positive integer n , then $x = 1$. (IMO shortlist 2012, N6)